Effects of Controls

AIM: Operate primary and ancillary controls in flight; observe and describe their effects.

- [ ] P **Checklists**: Student performs checklist with instructor assistance; no checklists while moving
- [ ] DP **Taxi**: Instructor demos basic taxiing, including parallax; student practices
- [ ] D **E8 Takeoff**: Instructor conducts; student follows through on ground roll, lift-off, and initial climb
- [ ] D **Stability**: Demo during transit — aircraft returns to level; student notes positive pitch stability
- [ ] D **Parallax**: Demonstrate horizon picture in left and right 30° bank
- [ ] DP **Elevator**: ±10° pitch — airspeed/RPM changes; ***EMPHASISE: pitch controls airspeed***; student practices
- [ ] DP **Ailerons**: 30° roll — patter slip, yaw, spiral tendency; student practices
- [ ] DP **Rudder**: Yaw — patter skid, roll, spiral tendency; student practices
- [ ] DP **Trim**: Demo pressure without trim; show correct direction; student practices
- [ ] DP **Power**: From slow flight, full power — pitch up, yaw/roll left; idle — pitch down, yaw/roll right; student practices
- [ ] DP **Flaps**: Stage by stage — pitch, increased drag, increased power; retract; student practices
- [ ] DP **Airspeed effect**: Student compares control feel at slow flight vs 110 kt
- [ ] D **Upset limits**: Safe pitch attitude limits; patter horizon reference
- [ ] P **Scenario**: "Using only further effects — pitch the nose up/down; roll left/right"

COMMON ERRORS

- **Overcontrolling** -> Wine glass grip, not beer mug
- **Fixating instruments** -> Horizon is primary; instruments confirm
- **Wrong trim direction** -> Pushing = finger down; pulling = thumb up
- **Missing further effects** -> "Feet follow ailerons" — never aileron alone
- **Poor lookout** -> Horizon, left, front, right, above/below

STANDARDS FOR PROGRESSION

P S

C NC

Progressing (P): Controls improving; lookout with prompting; names primary controls
Solo (S): Smooth inputs; lookout starting; names controls unprompted; IMSAFE verbally; >=1 TEM threat verbalised
NC: Persistent gross or opposite inputs

⚠️ **SAFETY** Exercises **above 1500 ft AGL** · Ceiling >2000 ft, wind <25 kt · Do not exceed V_{NO} · Pre-select forced landing area